



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

12.0 Industrial Wireless

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Understanding the basic principles, and applications of industrial wireless technologies and their role in enhancing operational efficiency and reducing costs.
- Identifying key wireless communication protocols, network structures and topologies in industrial wireless systems.
- Explain the importance of security and reliability with impact of industrial wireless on reducing greenhouse gas emissions and supporting sustainable practices.

12.0 Introduction to Industrial Wireless

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- How Industrial Wireless Contributes to Sustainability

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Case for Change for Industrial Wireless

1. Improved Operational Excellence

Smart Helmets

- Enables 'hands-free' operations for field workers in collaborative working environments.
- Features voice command capability for hands-free operation.
- Eliminates reliance on walkie-talkies or mobile devices for communication.

3. Enhanced Remote Operations

Industrial Internet of Things (IIoT)

- Improves connectivity of field devices.
- Reduces asset downtime with predictive and prescriptive maintenance

Drones

- Monitors infrastructure, equipment, safety, and security remotely.
- Enables aerial inspection, monitoring, and surveying of on-site operations

Robotics

- Supports closed inspection, monitoring, and surveying.
- Applies to pipelines, platforms, rigs, storage tanks, and other oil and gas structures.

2. Life Safety Solutions

Smart Mobile Devices (e.g., Phones/Tablets)

- Enhances a collaborative working environment for efficient wireless operations
- Allows capturing, recording, and uploading of data by providing real-time reports and dashboards.
- Supports managing operating procedures electronically and receiving real-time notifications.
- Supports remote monitoring and technical collaboration.

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What is Industrial Wireless?

- Industrial Wireless is the use of wireless communication technologies in industrial environments to enable real-time data exchange, monitoring, and control of machinery, sensors, and automation systems.
- Unlike traditional wired networks, industrial wireless systems provide **greater flexibility, scalability, and cost efficiency**, making them essential for modern manufacturing, process control, and industrial automation applications.
- Industrial wireless networks uses **various communication protocols** such as Wi-Fi, Bluetooth, Zigbee, LoRa, 5G, and private LTE to support different use cases.

What is Industrial Wireless - Key Advantages and Features of Industrial Wireless

No Wiring Required

Simplified Infrastructure

Fast Deployment

Flexible and Scalable

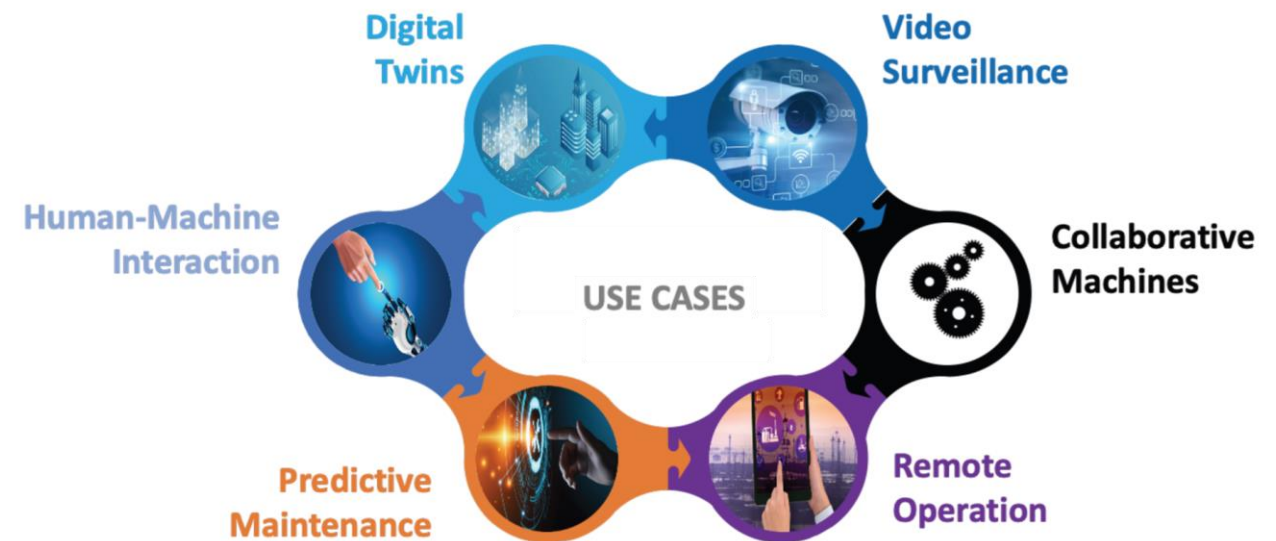
Enhanced Mobility

Reliable Communication

Lower Maintenance Costs

Improved Safety

Remote Monitoring



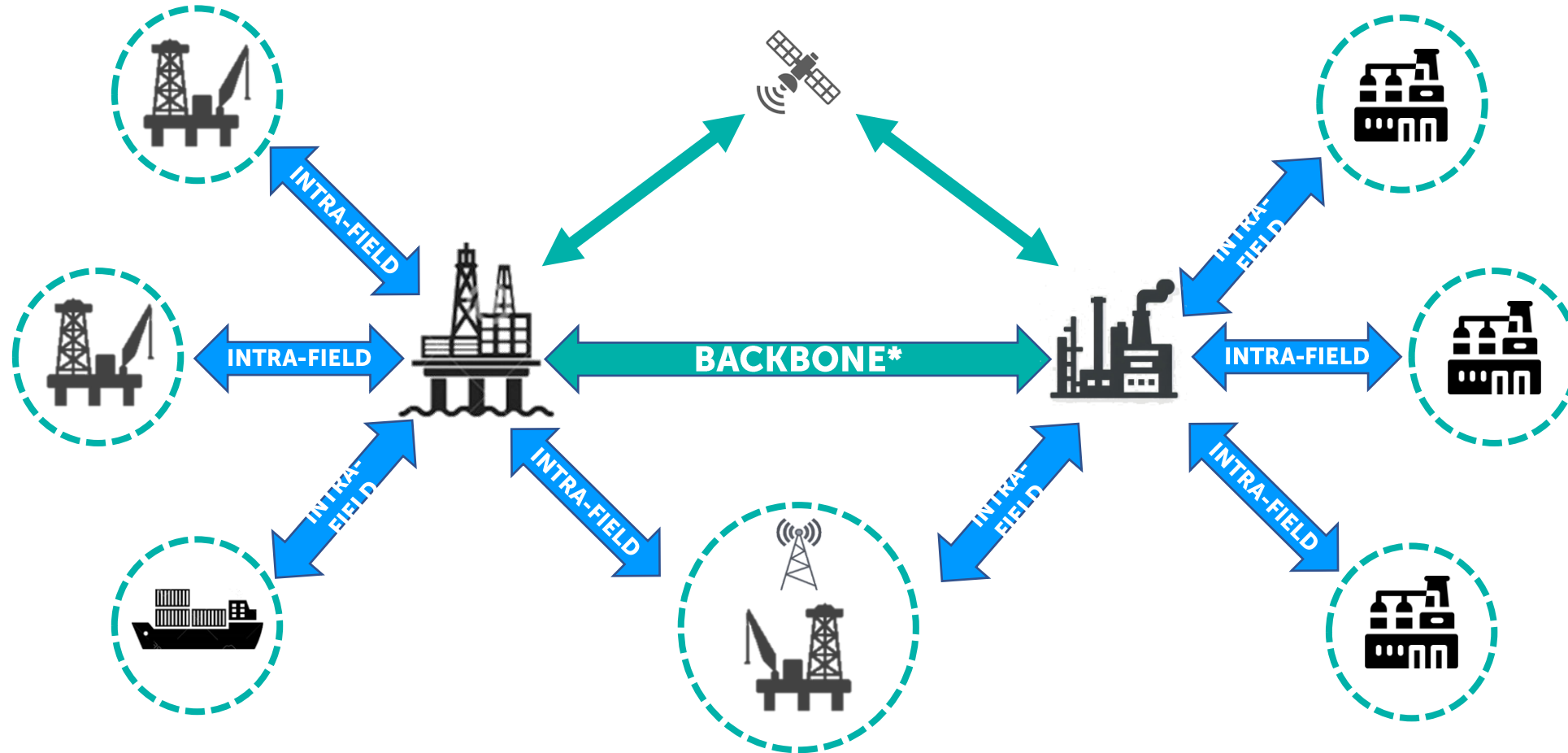
Source: https://www.mdpi.com/1424-8220/23/1/73?type=check_update&version=2

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Industrial Wireless in Telecommunication Infrastructure Hierarchy

Industrial Wireless as **Last Mile Connectivity** to provide network access to end/field-devices



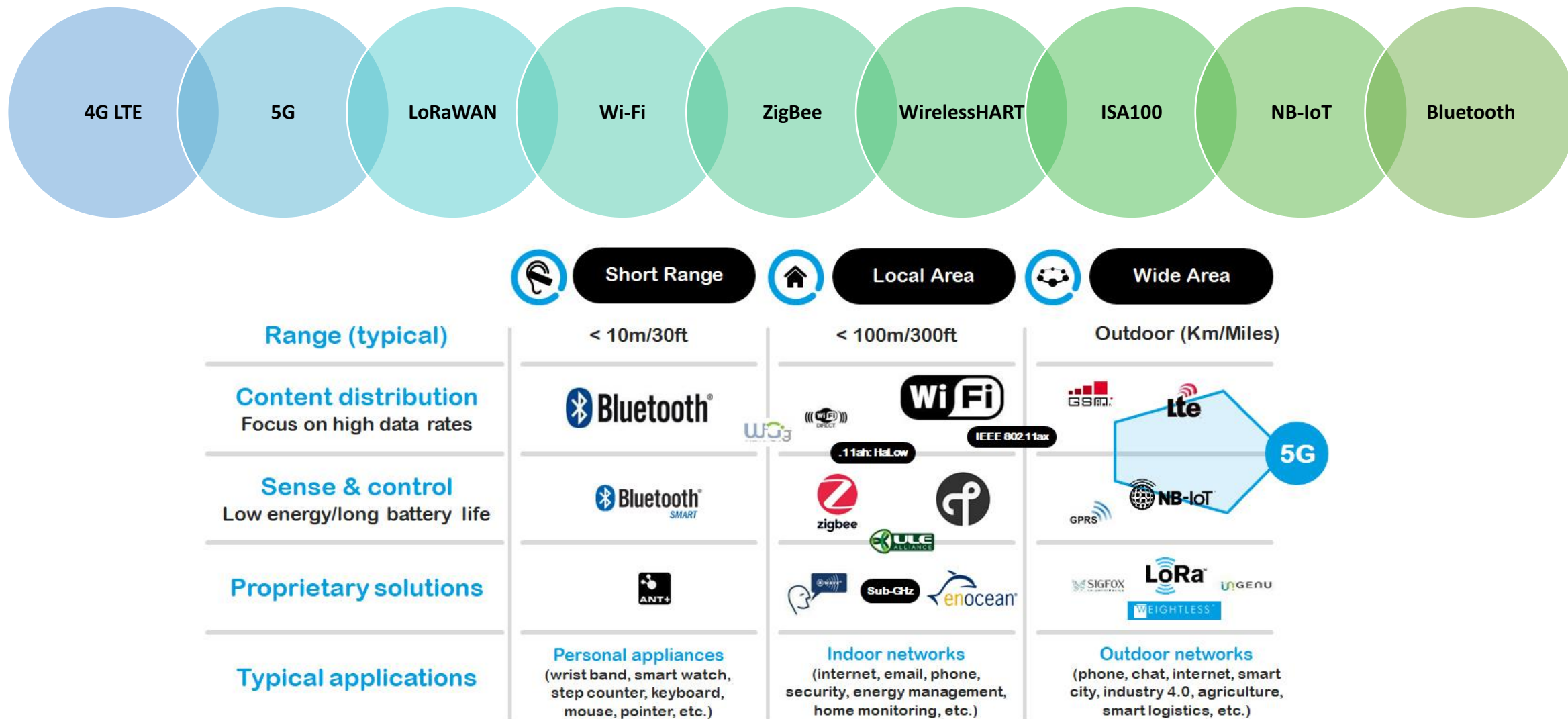
**Local Connectivity
a.k.a Last Mile**

**Sometimes uses term 'Inter-field' for offshore connectivity*

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Industrial Wireless Communication Protocols and Network Structure



Industrial Wireless Communication Protocols and Network Structure



1. 4th Generation Long Term Evolution (4G LTE) and 5th Generation (5G)

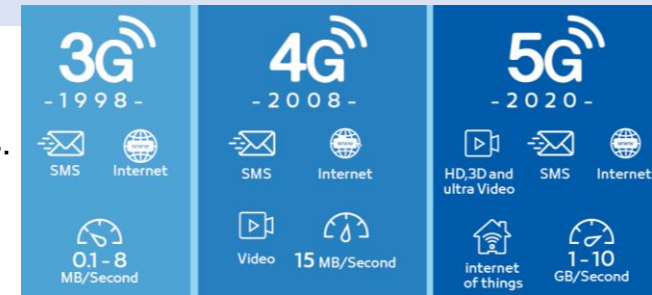
•**Description:** Cellular communication offers wide coverage, high data rates, and low latency, as defined by the 3rd Generation Partnership Project, a global consortium of seven telecommunications standard development organizations.

•**Features:**

- 4G: Reliable, moderate latency (~10ms), supports IoT applications requiring good throughput.
- 5G: Ultra-reliable low-latency communication (URLLC ~1ms), enhanced mobile broadband (eMBB), massive IoT device connectivity.

•**Network Structure:**

- Centralized cellular architecture with **Base Stations** connecting to core networks.
- Supports private networks (e.g. Private 5G, Campus Network).



•**Applications:** Remote monitoring, real-time control, AGVs, AR/VR for maintenance, IIoT devices, local backhaul in place of physical fibre optic cable to Substations/FARs/Metering Station etc.

Industrial Wireless Communication Protocols and Network Structure

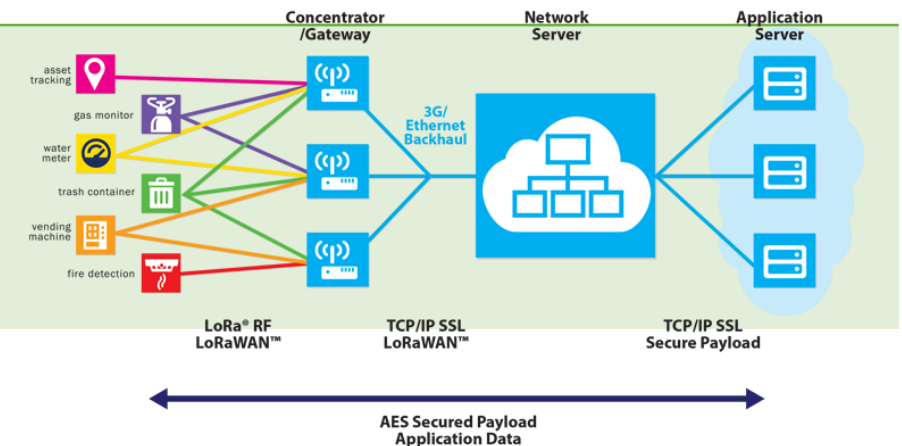


LoRaWAN (Long Range Wide Area Network)

•**Description:** A low-power, wide-area network (LPWAN) technology designed for IoT and sensor-based applications. It is developed by the LoRa Alliance, a non-profit organization, to wirelessly connect "things" to the internet in regional, national, or global networks.

•Features:

- Long-range (up to 15-20 km in rural areas).
- Low power consumption (battery life of devices >10 years).
- Low data rates (0.3 kbps to 50 kbps).



•Network Structure:

- **Star-of-stars topology:** End nodes (sensors) communicate with gateways that relay data to a centralized network server.

•**Applications:** Environmental monitoring, asset tracking, energy management, predictive maintenance.

Source: <https://loro-alliance.org/>

Industrial Wireless Communication Protocols and Network Structure



Wi-Fi (IEEE 802.11 Standards)

•**Description:** A ubiquitous wireless LAN standard offering high-speed data transfer over short to medium distances developed by Institute of Electrical and Electronics Engineers (IEEE), and certified for compliance by Wi-Fi Alliance, a non-profit organization.

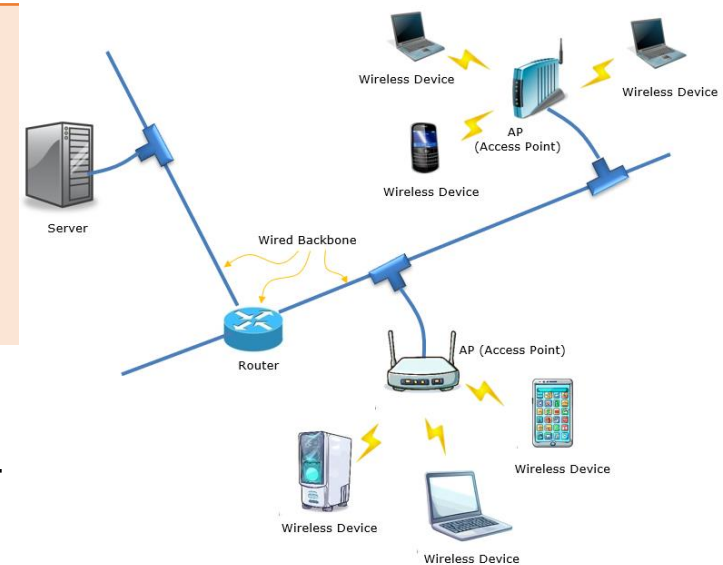
•Features:

- High throughput (up to several Gbps with Wi-Fi 6).
- Moderate latency.
- Limited range (~100m indoors, 300m outdoors).

•Network Structure:

- **Infrastructure mode:** Devices communicate via an access point (central hub).
- **Mesh topology:** Devices interconnect, forming a resilient network.

•**Applications:** Plant-floor connectivity, video surveillance, worker communication, and edge computing.



Industrial Wireless Communication Protocols and Network Structure



Zigbee

•**Description:** A short-range, low-power wireless protocol based on IEEE 802.15.4 for IoT and industrial applications. Developed by Connectivity Standards Alliance.

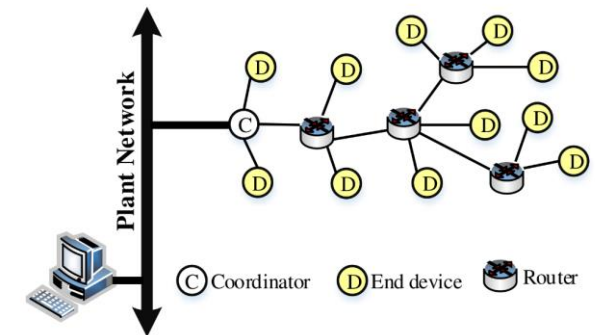
•**Features:**

- Operates at 2.4 GHz or 868/915 MHz.
- Low data rates (~250 kbps).
- Battery-efficient.

•**Network Structure:**

- **Mesh topology:** Devices communicate with each other, providing redundancy.
- A **coordinator** manages the network.

•**Applications:** Building automation, smart lighting, sensor networks in factories.



Source: <https://zigbeealliance.org/solutions/>
<https://commons.wikimedia.org/w/index.php?curid=106668572>

Industrial Wireless Communication Protocols and Network Structure

WirelessHART

WirelessHART

•**Description:** An industrial wireless protocol built on IEEE 802.15.4 and designed specifically for process industries. Developed by HART Communication Foundation.

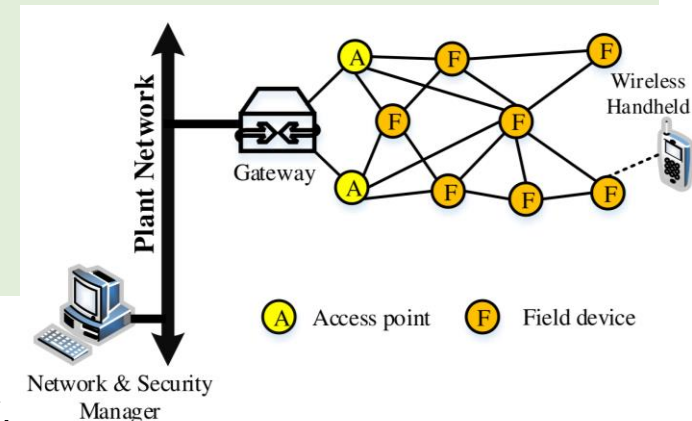
•**Features:**

- Time-synchronized communication.
- Reliable mesh networking with self-healing capabilities.
- Secure data transmission (AES encryption).

•**Network Structure:**

- **Mesh topology:** Devices (field instruments) form a robust, self-healing network.
- Managed by a **gateway** connecting to the control system.

•**Applications:** Process automation, environmental monitoring, hazardous-area sensors.



Source: <https://www.fieldcommgroup.org/>

Industrial Wireless Communication Protocols and Network Structure



ISA100.11a

•**Description:** A wireless standard for industrial automation and control, designed for high reliability and security developed by the International Society of Automation (ISA).

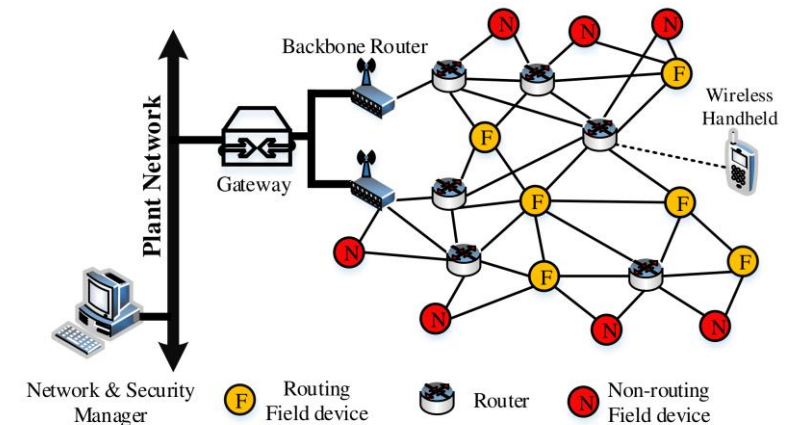
•**Features:**

- Operates in the 2.4 GHz band.
- Supports mesh, star, or hybrid topologies.
- Compatible with legacy systems.

•**Network Structure:**

- Flexible topology (mesh or star).
- Managed by a **gateway** connecting to control and enterprise systems.

•**Applications:** Wireless control, safety systems, and monitoring in industrial plants.



Industrial Wireless Communication Protocols and Network Structure



NB-IoT (Narrowband IoT)

•**Description:** A cellular LPWAN technology optimized for IoT devices developed by 3GPP. It operates over existing mobile network.

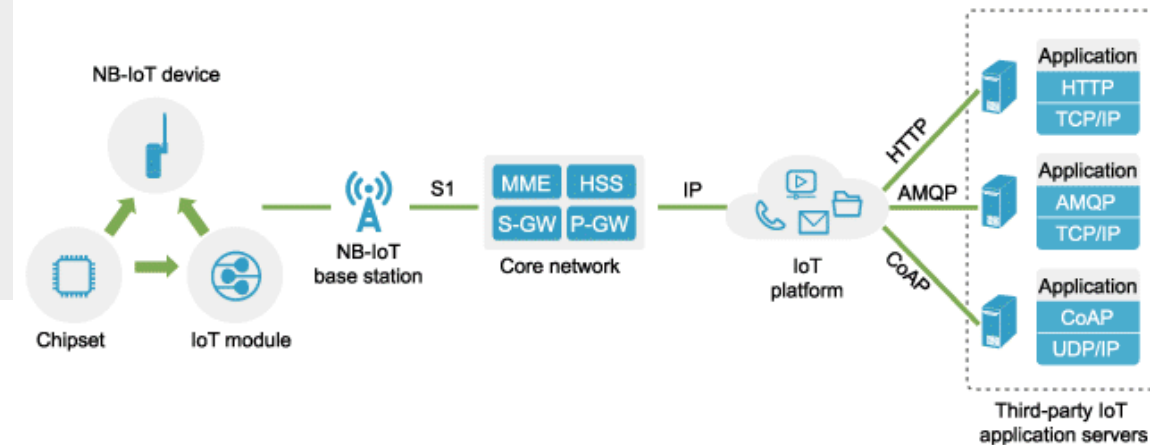
•**Features:**

- Wide coverage, including deep indoor environments.
- Low power consumption (battery life >2 years).
- Low data rates (max ~250 kbps).

•**Network Structure:**

- Part of the existing cellular network (star topology).
- Direct communication between IoT devices and cellular base stations.

•**Applications:** Smart metering, environmental monitoring, low-data industrial IoT use cases.



Source: <http://www.gsma.com>

Industrial Wireless Communication Protocols and Network Structure



Bluetooth (including BLE)

•**Description:** Short-range wireless communication technology commonly used for personal devices and industrial sensors developed and licensed by Bluetooth Special Interest Group (SIG), a non-profit organization.

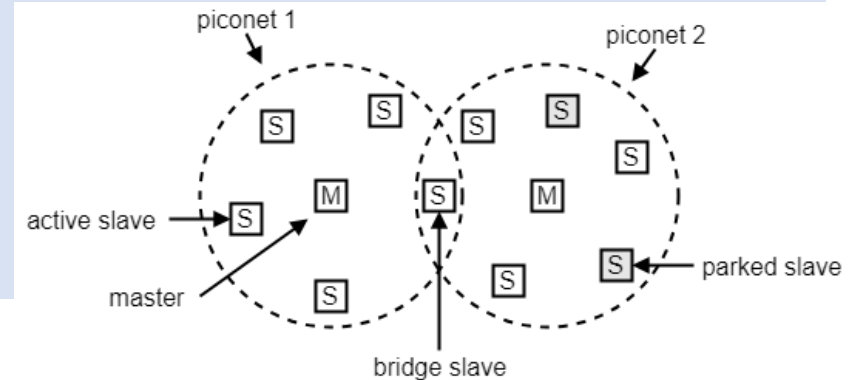
•Features:

- Low power (especially Bluetooth Low Energy, BLE).
- Range: ~10-100m.
- Moderate data rates (up to 2 Mbps for BLE).

•Network Structure:

- **Star topology:** Central device communicates with multiple peripherals.

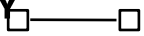
•**Applications:** Wearables, portable diagnostic tools, and proximity-based sensors.



Source: <http://www.bluetooth.com>

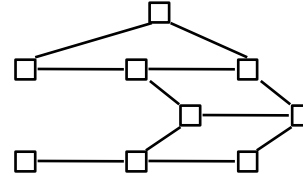
Industrial Wireless Communication Protocols and Network Structure

POINT-TO-POINT TOPOLOGY



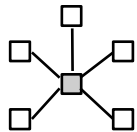
(Between 2 units in communication)

GRID TOPOLOGY



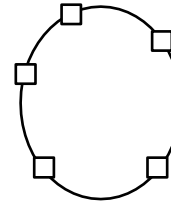
(Devices are linked to one another, forming a "spider's web". There are a number of possible paths for reaching a node)

STAR TOPOLOGY



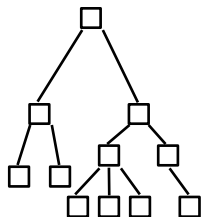
(Several units communicating via their own line with a Central unit)

RING TOPOLOGY



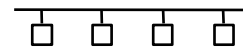
(All the units are connected in series in a closed loop.
☒ Communications must pass via all the units to arrive at the receiver)

TREE TOPOLOGY



(This is a variant of the star topology)

BUS TOPOLOGY



(The network consists of a main line to which all the units are connected)

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Wireless Communication Protocols: Advantages and Limitations

Protocol	Range	Data Rate	Topology	Power Consumption	Use Case Examples
4G LTE	Wide (<5km)	Medium high (<200Mbps)	Star	Medium high (0.5W - 5W)	Remote monitoring, mobile asset tracking, and real-time data collection for dispersed assets.
5G	Wide (<5km)	High (Gbps)	Star	Medium to High (1W - 10W) ^[1]	Ultra-low-latency use cases like robotics, autonomous vehicles, AR/VR for maintenance, and high-bandwidth streaming.
LoRaWAN	Long (>5km)	Low (50 kbps max)	Star-of-stars	Very Low (1 mW - 100 mW)	Long-range, low-power IoT applications like utility metering, remote sensors, and environmental monitoring.
Wi-Fi	Short (~100m)	Medium to High (Mbps-Gbps)	Star/Mesh	Medium (100 mW - 500 mW)	Worker communication, plant-floor devices, video surveillance, and edge computing.
Zigbee	Short (~100m)	Low (250 kbps)	Mesh	Low (10 mW - 100 mW)	Localized industrial sensor networks, smart lighting, and short-range monitoring/control systems.
WirelessHART	Medium (~100m)	Low (250 kbps)	Mesh	Low (10 mW - 100 mW)	Process automation, hazardous area monitoring, and predictive maintenance in industrial plants.
ISA100	Medium (~100m)	Low (250 kbps)	Mesh/Star	Low (10 mW - 100 mW)	High-reliability applications in safety-critical systems and hybrid automation environments.
NB-IoT	Wide (km)	Low (250 kbps max)	Star	Very Low (1 mW - 100 mW)	Deep indoor connectivity, smart meters, remote sensors, and low-bandwidth industrial IoT.
Bluetooth	Short (~10m)	Medium low (2 Mbps)	Star	Very Low (1 mW - 50 mW)	Short-range applications like wearables, portable diagnostic tools, and proximity-based sensors.

Wireless Communication Protocols: Advantages and Limitations

[\[Open\]](#)

When to Choose a Protocol Based on Cost vs Usage

- **Low-cost protocols** like Zigbee and LoRaWAN are ideal for non-critical IoT applications where power efficiency and range are prioritized over speed.
- **Moderate-cost protocols** like WirelessHART are preferred in industrial settings where robust, reliable, and secure communication is needed.
- **High-cost protocols** like Private 5G Network are reserved for industries where reliability, security, performance, low latency, and high bandwidth outweigh budget constraints, such as digital programs in process areas, aerospace, autonomous robotics, or large-scale manufacturing.

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How Industrial Wireless Contributes to Sustainability

Lower Carbon Emissions from Cable Manufacturing

- Avoid copper and plastic production, reducing CO₂ by up to 3.5 tons per ton of cable.

Reduced Installation Emissions

- No heavy equipment or materials needed for excavation and laying cables (e.g., conduits, cable trays).

Longer Lifespan, Less Waste

- Wireless systems need fewer physical upgrades (e.g., cable upgrade), cutting e-waste.

Indirect Energy Savings

- Utilize more devices for real-time monitoring and automation which prevent the usage of more cables and the device only transmit data when required (Device in sleep mode consumes power when needed) intern reduces energy consumption.

Group Discussion

Group Discussion on Industrial Wireless applications e.g., Case Study on reliability improvement through IIoT deployment using Wireless HART protocol

